

The Worldsheet and the Copper Plane

A string scattering amplitude is the thermal-noise statistics of a resistive sheet

Model an interacting particle not as a zero-dimensional point but as an extended one-dimensional string, sweeping out a continuous two-dimensional worldsheet in spacetime. Its quantum probability amplitude is then, integrand for integrand, the same mathematical object as the statistics of thermal fluctuations of electrical power across a two-dimensional resistive copper plane. In the correspondence the momenta of the incoming and outgoing particles become discrete electrical currents, injected and extracted at coordinate locations z_i strictly along the one-dimensional outer boundary of the board. A

two-dimensional sheet has a logarithmic Green's function, so the power dissipated by any Kirchhoff-balanced set of edge currents is a sum of pairwise logarithms of the contact separations; and Johnson–Nyquist noise weights each microscopic thermal state by the exponential of its dissipated heat. Integrating those thermal states by sliding the injection points continuously along the boundary therefore reproduces the quantum path integral of the interacting strings, governed by the mathematical equivalence:

$$\int e^{-P/P_N} dz_1 \cdots dz_N = \int \prod_{1 \leq i < j \leq N} |z_i - z_j|^{2\alpha' p_i p_j} dz_1 \cdots dz_N$$

READ AS A COPPER PLANE

the statistical weight of the dissipated power

P dissipated power P_N the Nyquist power scale (p.3) α' the Regge slope (p.3) N contacts
an identity between INTEGRANDS; only $N-3$ of the N positions may actually be integrated (p. 4)

READ AS A STRING WORLD SHEET

the Koba–Nielsen factor of the N -point amplitude

NOT A NEW OBSERVATION — THE CHAIN OF RESULTS IT RESTS ON

1730 Euler

the integral interpolating the factorial, to Goldbach, 8 January.

1928 Johnson · Nyquist

measure, then explain, resistor noise. Phys. Rev. 32, 97 & 110.

1968 Veneziano

the crossing-symmetric four-point amplitude. Nuovo Cimento A57, 190.

1969 Koba · Nielsen

generalise it to N particles — the product above. Nucl. Phys. B10, 633.

1970 Fairlie · Nielsen

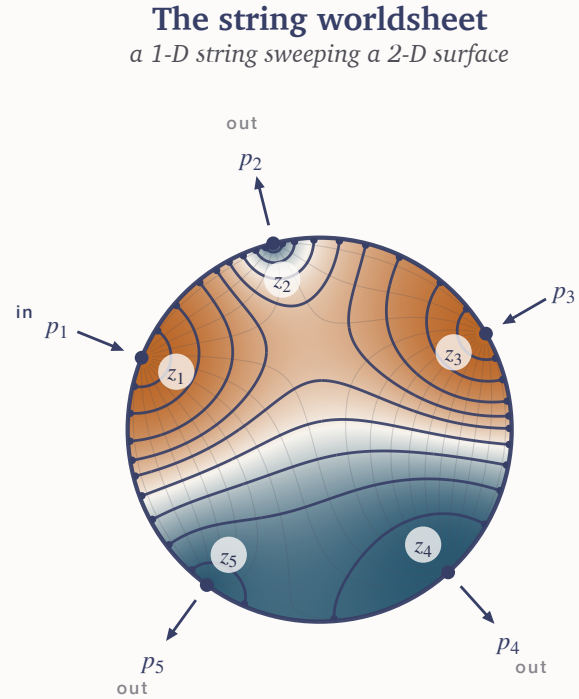
the analogue model — the two sides joined. Nucl. Phys. B20, 637.

1981 Polyakov

makes the worldsheet a 2-D statistical system. Phys. Lett. B103, 207.

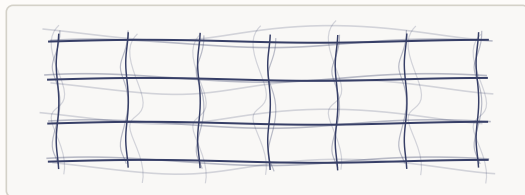
One integrand, two readings

The same harmonic field on the same disk, with the same five boundary points



$$\langle X^\mu(z_i) X^\nu(z_j) \rangle = -2\alpha' \eta^{\mu\nu} \ln |z_i - z_j|$$

τ runs copper $\rightarrow$ teal, the way the current flows

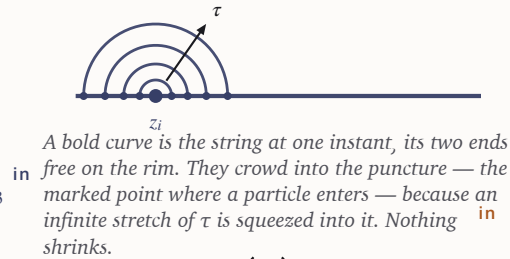


Microscopic fluctuations: quantum

The embedding X^μ fluctuates; each history weighted $\exp(-S)$.

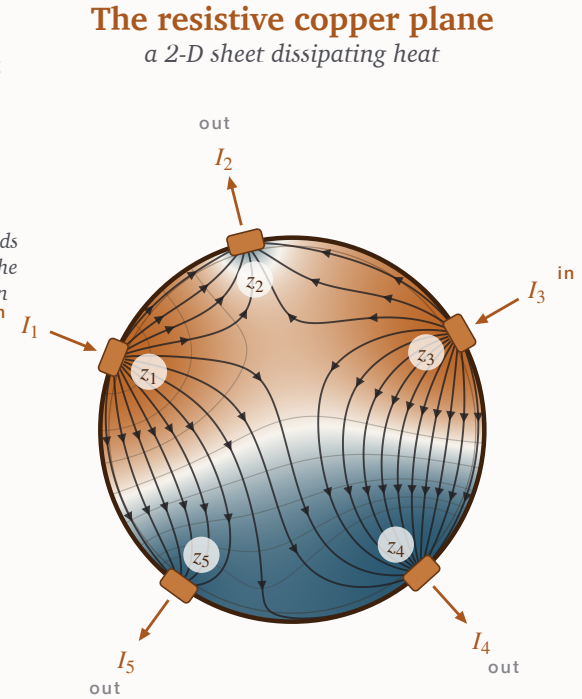
Read each current as a light-cone momentum; Mandelstam's map $\rho = \sum_i \alpha_i \ln(z - \zeta_i)$ gives $\tau = -(\pi/R_s)V$, so the voltage IS the string's time, and $\sigma = \text{Im} \rho$ is position along it.

DETAIL — THE RIM AT ONE PUNCTURE



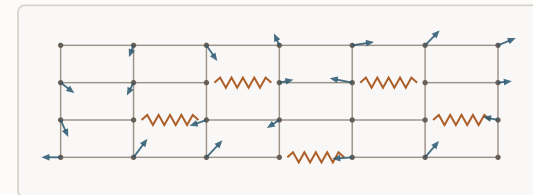
ONE INTEGRAND TWO READINGS

$$\begin{aligned} p_i &\leftrightarrow I_i \\ \tau &\leftrightarrow -(\pi/R_s)V \\ \alpha' &\leftrightarrow R_s/\pi P_N \\ e^{-S} &\leftrightarrow e^{-P/P_N} \\ \Sigma p_i = 0 &\leftrightarrow \Sigma I_i = 0 \end{aligned}$$



$$V(w) = -(R_s/\pi) \sum_i I_i \ln |w - \zeta_i|$$

the potential — exact, because the currents sum to zero



Microscopic fluctuations: thermal

Every element rattles; each pattern weighted $\exp(-P/P_N)$.

Colour is that same field on both sides — copper positive, teal negative. The flow lines are the current density $J = -\nabla V/R_s$; they are also the lines of constant σ .

Five steps, each one measured

Every ✓ below is a number produced by `verify_string_copper.py`, not a citation

THE CHAIN, IN ORDER

1 The string side is a product of powers.

An open string's vertex operators — the insertions standing for each incoming or outgoing particle — sit on the one-dimensional boundary of the worldsheet, where the embedding has the propagator $\langle X^\mu(x)X^\nu(y) \rangle = -2\alpha' \eta^{\mu\nu} \ln|x-y|$. Contracting N of them turns that logarithm into a product, because the Gaussian average of a product of exponentials is the exponential of the sum of pairwise propagators:

$$\prod_{i<j} |z_i - z_j|^{2\alpha' p_i p_j}$$

This is the bosonic string, and its $N = 0$ state is the tachyon ($m^2 = -1/\alpha'$, the bottom of the ladder in consequence 3).

✓ the product is exp of a logarithmic pair energy, to machine precision

2 A sheet's rim Green's function is $-R_s/\pi$ times a log.

A two-dimensional sheet has a logarithmic Green's function, and a contact ON an insulating rim sees twice the interior value, because its image charge sits on top of it. That factor of two is not bookkeeping: it is the same two that separates the open-string boundary propagator from the closed-string bulk one.

✓ measured -0.317971 against $-R_s/\pi = -0.318310$, 0.11%; the error falls by $4.0\times$ per doubling, i.e. second order

3 Neutral rim currents dissipate a sum of logs.

Feeding that into $P = \sum_i I_i V(z_i)$ gives

$$P = -(2R_s/\pi) \sum_{i<j} I_i I_j \ln|z_i - z_j| + \kappa \sum_i I_i^2$$

with κ the self-energy coefficient, set by a pad's finite radius. There is no factor of one half here — that belongs to the energy of assembling static charges, not to the power delivered by current sources, and inserting it would halve α' .

✓ fitted 0.637208 against $2R_s/\pi = 0.636620$ over 300 random neutral patterns, $R^2 = 0.9999992$

AND ITS TWO CLOSING MOVES

4 Nyquist supplies the exponential — and its scale.

Thermal noise makes the contact currents a Gaussian whose quadratic form is exactly the dissipated power, so a fluctuation's weight is $\exp(-P/P_N)$ with $P_N = 8k_B T \Delta f$ for a noise bandwidth Δf (equivalently $4k_B T/t$ for an averaging window t). Writing $\exp(-P)$ alone is dimensionally incomplete: a power is not an exponent. Exponentiating step 3 now turns the sum of logs into a product of powers —

$$e^{-P/P_N} = \prod_{i<j} |z_i - z_j|^{(2R_s/\pi P_N) I_i I_j}$$

— and matching that exponent against $2\alpha' p_i p_j$ gives $\alpha' = R_s/\pi P_N$: the Regge slope, the constant fixing the string's mass spectrum, as a sheet resistance in units of noise power. Its units are one over current squared, so $\alpha' p^2$ is dimensionless.

✓ the substitution measured end to end on 6 configurations — worst implied error on P 0.11%, which is the lattice error of step 2

5 Gauge-fix three contacts, then slide the rest.

Fixing $z_1, z_3, z_4 = 0, 1, \infty$ and integrating z_2 over $(0, 1)$ collapses the $N = 4$ product to Euler's Beta integral, with $\alpha(s) = 1 + \alpha's$ and s, t the Mandelstam invariants — the squared collision energy and momentum transfer:

$$\int_0^1 x^{-\alpha(s)-1} (1-x)^{-\alpha(t)-1} dx = B(-\alpha(s), -\alpha(t)) = \Gamma(-\alpha(s))\Gamma(-\alpha(t)) / \Gamma(-\alpha(s)-\alpha(t))$$

where $B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx$, and Γ is the integral Euler sent Goldbach in 1730 to interpolate the factorial. That is the Veneziano amplitude.

✓ the integral equals Euler's B to 4.4×10^{-16}

Three consequences, and the evidence

Each consequence is precise rather than poetic; each number below is produced by `verify_string_copper.py`

The three orderings are the three channels.

Four contacts admit three distinct cyclic orders around the rim. Those are the three integration ranges of x , and they give $B(-\alpha(s), -\alpha(t))$, $B(-\alpha(t), -\alpha(u))$ and $B(-\alpha(s), -\alpha(u))$ — the s-t, t-u and s-u channels, whose sum is the crossing-symmetric total.

✓ all three verified to 1.9×10^{-10} or better

The Beta function interpolates a binomial.

$B(k+1, n-k+1) = 1/[(n+1)C(n,k)]$ exactly — so B is the continuous version of a combinatorial quantity, though it enters as the reciprocal. That is a separate fact from the three orderings, and not the reason there are three.

✓ 38 integer pairs, 0 mismatches

The particle spectrum is two pads touching.

The poles come from the endpoint $x \rightarrow 0$, where two contacts merge. In the Γ form they are visible directly: $\Gamma(-\alpha(s))$ blows up at $\alpha(s) = 0, 1, 2, \dots$, i.e. $\alpha's = -1, 0, 1, \dots$, which is the open-string spectrum $m^2 = (J-1)/\alpha'$.

✓ $aB(a, 1.5) \rightarrow 1$ as $a \rightarrow 0$: 0.854, 0.944, 0.982, 0.994

WHAT WAS MEASURED

the sheet is a lattice of unit resistors solved exactly by DCT; the washer row is a conjugate-gradient solve

QUANTITY	MEASURED	EXACT	AGREEMENT
Rim-contact pair potential, $R_s = 1$	−0.317971	$-R_s/\pi = -0.318310$	0.11%
Interior-contact pair potential	−0.159233	$-R_s/2\pi = -0.159155$	0.05%
Rim / interior ratio	1.99689	2 exactly	the image charge
Power-law coefficient, 300 patterns	0.637208	$2R_s/\pi = 0.636620$	0.09%, $R^2 = 0.99999992$
Non-neutral ΔP per size doubling	0.8825	$(Q^2 R_s/\pi) \ln 2 = 0.8825$	0.01%
Gauge-fixed integral vs Euler B	0.26666666666667	$B(2, 1.5) = 0.26666666666667$	4.4×10^{-16}
$SL(2, \mathbb{R})$ measure ratio, on the mass shell	1.000000000000	1 exactly	four Möbius maps
the same, pushed off the mass shell	0.065232	$\neq 1$	the symmetry is the shell
Koba–Nielsen exponent eigenvalues	−3, 0, 5, 6	indefinite, rank 3	a real current gives rank 1, ≥ 0
Washer resistance, $a = 120$, $b = 600$	0.256748	$(1/2\pi) \ln(b/a) = 0.256150$	0.23%, first order in h

Euler to Goldbach, 8 Jan 1730 · J. B. Johnson, Phys. Rev. 32 (1928) 97 · H. Nyquist, Phys. Rev. 32 (1928) 110 · G. Veneziano, Nuovo Cimento A57 (1968) 190 · Z. Koba & H. B. Nielsen, Nucl. Phys. B10 (1969) 633 · D. B. Fairlie & H. B. Nielsen, Nucl. Phys. B20 (1970) 637 · A. M. Polyakov, Phys. Lett. B103 (1981) 207

Six things the one-line version hides

None of these breaks the identity; each one bounds what it may be read to mean

1 One plane carries one spacetime dimension.

The obstruction is RANK, not sign. $2\alpha' p_i p_j$ has rank up to $D-1$ — rank 3 at the kinematic point $s = -3, t = -2.5, \alpha' = 1$, where its eigenvalues are $-3, 0, 5, 6$ — while a real scalar current can only produce $I_i I_j$, which is rank ONE. (Indefiniteness is not the reason, and is not even general: at $s = -2, t = -1.5$ the same matrix is positive semi-definite.) One copper plane therefore realises a single target dimension; D dimensions need D superposed planes, the timelike one carrying imaginary current. Reading a current as a momentum vector is the one step in the statement that cannot be taken literally.

2 The integral as written is still divergent.

The measure is exactly $SL(2, \mathbb{R})$ -invariant — measured ratio 1.000000000000 under four Möbius maps — and it needs BOTH conditions: the exponent collected at puncture i is $2\alpha' p_i^2 - 2 - 2\alpha' p_i \cdot (\Sigma p)$. Breaking the mass shell alone gives 0.0652; breaking momentum conservation alone gives 0.0362. Given $\Sigma p = 0$ it is $\alpha' p^2 = 1$ that does the work — that condition is exactly “the vertex operator has conformal weight one”. Either way $\int dz_1 \dots dz_N$ carries the infinite volume of the gauge group, so three punctures must be held fixed and only $N-3$ integrated: an identity between integrands first, and between amplitudes only after.

3 The self-energy is a regulator, not the mass shell.

The $i = j$ terms diverge on both sides: on the sheet the cut-off is a pad’s finite radius, on the worldsheet it is normal ordering. But they are not the same object. $\kappa \Sigma I_i^2$ is a multiplicative constant, per plane; $\alpha' p^2 = 1$ is a CONDITION on the full D -dimensional momentum, and no single plane can enforce a constraint that sums over D of them. The two are related regularisations, not two names for one thing.

4 Rim versus interior is open versus closed.

The factor of two between a contact on the insulating rim and one in the interior — measured as 1.99689 — is the same factor of two between the open-string boundary propagator $-2\alpha' \ln|x-y|$ and the closed-string bulk one $-\alpha' \ln|z-w|$. Moving a contact off the edge into the copper is not a small perturbation of the model: it converts an open-string insertion into a closed-string one.

5 This is the worldsheet CFT, and not a least-resistance path.

The two-dimensional theory here is the string’s own worldsheet, and the copper plane is that worldsheet — not spacetime. It is not the boundary CFT of AdS/CFT, where the dual theory lives on the boundary of the spacetime the gravity fills; conflating those is the most inviting misreading of the statement. Nor is the amplitude a least-resistance path — though not for the obvious reason. Rayleigh’s minimum is exactly what sits in the exponent: for a given set of contact currents the weight is $\exp(-P/P_N)$ with P the OHMIC dissipation, the fluctuations about it contributing only a source-independent factor that cancels. What the amplitude then does is integrate over the contact POSITIONS. Neither step is a minimisation over paths.

6 The two readings use different charges.

The strings on page 2 come from Mandelstam’s map, which reads each current as a light-cone momentum p^+ — one component. The Koba–Nielsen weight reads the same current as the whole momentum vector. Two identifications, not one.

Punch a hole in the copper

Two routes, and both of them are geometry

Everything so far has been OPEN strings: contacts on the rim, insertions on the boundary. Their lightest massless state is spin 1 — a gauge boson, like a photon — so there is no gravity anywhere in it. Gravity is carried by CLOSED strings, whose lightest massless state is spin 2; and a massless spin-2 field has no choice but to couple to everything’s stress-energy, which is what gravity is. On a worldsheet, open versus closed is literally rim versus interior. So there are exactly two ways to put gravity into a sheet of copper, and both are changes to its GEOMETRY rather than new ingredients.

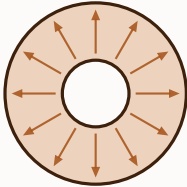
ROUTE ONE — MOVE A CONTACT OFF THE RIM

	A CONTACT ON THE RIM	ONE IN THE INTERIOR
the insertion sits	on the 1-D rim	in the 2-D interior
so the string is	open	closed
its propagator	$-2\alpha' \ln x-y $	$-\alpha' \ln z-w $
lightest massless state	spin 1 — a gauge boson	spin 2 — the graviton

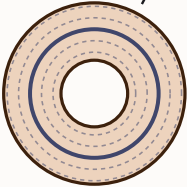
The factor of two measured between those two propagators — 1.99689 — is that divide showing up in a lattice Green’s function. But a current injection is the WRONG KIND of insertion. $e^{ip\cdot X}$ is a monopole, and the state it makes is the tachyon. A graviton operator $\epsilon_{\mu\nu} \partial X^\mu \partial X^\nu e^{ip\cdot X}$ carries that same monopole factor — it injects current like anything else — but multiplied by a factor bilinear in ∂X , which is the current density. So route one says WHERE gravity lives, not how to source it.

ROUTE TWO — PUNCH A HOLE

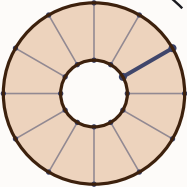
Change the sheet’s TOPOLOGY instead. Copper with a hole has two rims, and in string perturbation theory each hole is one LOOP — so a washer is the one-loop diagram. Cut it open and roll it up and it is a cylinder of length $\ln(b/a)$. It can be sliced two ways, exactly as page 2 sliced the disk, and neither is more correct:



as a resistor
radial flow, radius a out to b



sliced by circles
each slice a CLOSED string, rim to rim



sliced radially
each slice an OPEN string, once round

Both describe the same washer — which is the whole point, and page 7 takes it up. Conformally that cylinder’s length is $\ln(b/a) = 2\pi R/R_s$, so resistance is SEPARATION between the rims. Raise it and the closed string travels far, heavy states die off exponentially and only the lightest survive the trip; lower it toward $b/a \rightarrow 1$ and the open-string loop takes over instead.

Three things it says gravity is

and one thing it cannot say

1 Gravity is not optional.

One washer, two slicings: the open-string loop and the closed-string exchange are the SAME object, so you cannot have the first without the second. A theory of open strings — of gauge fields on a boundary — is not consistent without closed strings in its interior. On this reading gravity is not an extra force anyone put in; it is a consistency requirement of the gauge sector. The copper picture makes that visible: you cannot slice a washer only one way.

2 Gravity reads the flow’s anisotropy, not its strength.

Across D superposed planes the graviton’s bilinear $\partial X^\mu \partial X^\nu$ is $J^\mu J^\nu$, a matrix of per-plane current densities, and it splits in two. The TRACE, $\Sigma_\mu (J^\mu)^2$, is the total Joule dissipation, and that part belongs to the DILATON. The TRACELESS remainder — how the dissipation is distributed among the planes — is the graviton. So gravity couples to the anisotropy of the flow rather than to how much heat is being made, and it couples wherever ANY current flows, whichever contact it came from. That is why gravity is universal. It also shows why a graviton needs at least two planes: a symmetric traceless tensor in one dimension is identically zero.

3 Its long range is the high-resistance end.

The amplitude integrates over the washer’s shape, and the far end of that range — where the ratio and the resistance both run to infinity and the cylinder is long — is where only the lightest closed states survive the crossing. That limit is the long-distance force. Gravity’s reach is not a separate fact about gravity; it is which end of the shape integral you happen to stand at.

What a “modulus” is, and why it is here.

A conformal map may stretch and bend a shape but never shear it. Any washer maps onto any other with the same ratio b/a — and onto NONE with a different one — so a washer has exactly one shape parameter conformal maps cannot remove. That leftover number is its modulus, $t = (1/2\pi)\ln(b/a)$, and an amplitude must integrate over it because it sums over every shape the surface can take: page 3’s gauge-fixing, one loop up. Measured, $a = 120$ and $b = 600$ give $R = 0.256748$ against $t = 0.256150$, 0.23% — so an ohmmeter on a washer reads a coordinate on the space of shapes.

What it cannot say.

This is the BOSONIC string, and that matters here. Its lightest closed state is not the graviton but the closed TACHYON, and the massless level holds a dilaton and a B-field beside the graviton — so the long-cylinder limit is not graviton exchange alone, nor even dominated by it. More basically: the copper is the WORLDSHEET, so gravity arrives as a string STATE, a particle that gets exchanged. The analogy gives the particle-physics face of gravity and nothing of the geometric one — no metric, no curvature, no Einstein equation.

READING ALL OF THIS CORRECTLY

EXACT — AN IDENTITY

The integrand identity; the rim-versus-interior factor of two; R/R_s as the annulus modulus. Theorems about one body of mathematics, measured here.

ANALOGY — A TRANSLATION

The page-2 dictionary translates; it does not identify. A current is one number, a momentum a spacetime vector — one plane, one dimension (p. 5).

NOT CLAIMED

That a board gravitates beyond the pull of its mass; that measuring copper could test string theory; or that this is holography (p. 5).